

Philopateer Boctor

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EDUCATION

Yale University

New Haven, CT

Bachelor of Science in Mechanical Engineering (ABET). GPA: 3.5

Aug. 2022 – May 2026

Relevant Coursework: Mechanical Design; Machine Elements; Strength of Materials; Solid Mechanics Lab; Dynamics; Thermodynamics; Fluid Mechanics; Soft Robotics; Computer-Aided Engineering; Materials Science, Mechatronics

EXPERIENCE

Verustruct

June 2025 – Aug 2025

Mechanical Engineering Intern

New Haven, CT

- Owned the mechanical architecture for both the horizontal and vertical retraction/extension mechanisms of Verustruct's multi-directional Translational Slip-Form (TSF) system, creating the first integrated, full-scale proof-of-concept adopted for the metal prototype now being manufactured at the HAX accelerator.
- Designed the subsystem to meet strict packaging, weight, and manufacturability requirements, creating a compact slipform envelope that fits one-to-two person lift limits and preserves internal volume for pumps, sensors, and future robotic subsystems.
- Improved the slipform's retraction reliability by analyzing binding events in full-scale prototypes and adjusting member stiffness and clearances, which prevented jamming and allowed the mechanism to cycle smoothly under actuator load.
- Owned concept generation and detailed NX development for the TSF retraction mechanisms and walked the founder through weekly design reviews, shaping decisions on the subsystem's final geometry, assembly strategy, and integration envelope for the company's flagship construction robot.

Yale University GRAB Lab

Oct. 2023 – May 2025

Undergraduate Researcher | PI: Prof. Aaron Dollar

New Haven, CT

- Improved the GRAB Lab's underactuated Stewart Hand by replacing its pulley-based 1-to-3 differential with a compact gear-driven design that enhanced planar positioning accuracy and provided more consistent torque distribution to all three fingers.
- Built the complete multi-stage drivetrain and iterated custom miter gears and tolerances until a single motor input produced coordinated motion in all three fingers, demonstrating functional grasping on objects like a Rubik's cube and establishing the mechanical baseline for the next control-integration phase.

PROJECTS

Translational Slipform Feed System | Capstone Project

Aug 2025 – Present

- Leading a team to develop Verustruct's material feed system, transforming pumpable concrete into a zero-slump mix for a translational slipform robot.
- Designed and fully CAded a conveyor-driven test platform that mimics robot motion, enabling repeatable, system-level extrusion testing under realistic conditions.
- Integrated sensing via a LabJack DAQ (temperature, humidity, VMA flow rate, conveyor speed) to enable quantitative analysis of process parameters and material behavior.
- Identified rapid segregation in early tests, driving adoption of continuous in-line mixing and informing subsequent design iterations.

In-Pipe Soft Robot for Locomotion

Jan 2025 – May 2025

- Led a team of three graduate students to design and fabricate a McKibben-actuated soft crawler, developing mold-ready CAD and custom silicone actuators inspired by research literature.
- Developed actuator and mold designs with alignment and wall-thickness control features to reduce rupture risk and improve reliability, enabling successful straight-pipe locomotion in the final demonstration.

TECHNICAL SKILLS

Software & Programming: SolidWorks, NX, Solid Edge, Onshape, MATLAB, Intelli-MAX, Arduino, L^AT_EX

Design & Analysis: 3D Design and Rapid Prototyping (FDM/SLA), Finite Element Analysis (FEA), GD&T

Machining & Fabrication: Proficient with lathe, mill, waterjet, laser cutter, and common machine shop tools